Chapter Two

IoT andM2M

IoT levels and Development Template

Based upon the number of monitoring nodes used, type of data base used, complexity/ simplicity of analysis, computation there **are 6 levels of IoT**. Different applications are implemented based on this level.

The IoT systems consist of these following components.

- Device
- Resources
- Controller Service
- Database
- Web Service
- Analysis Component
- Application Device:

Device: The IoT device allows identification, remote sensing, actuating, and remote monitoring capabilities

Resource: Resources are the software components on the IoT device for accessing, processing and storing sensor information, or controlling actuators connected to the device Resources include the software components that enable network access for the device. For ex: The programs that we have written for object detection using IR sensors, to find out the distance using ultra sonic sensor etc.

Controller Service: Controller service is a native service that runs on the device and interacts with the web services. Controller service sends data from the device to the web service and receives commands from the application for controlling the web services. For ex: The ESP 8266 programming, setting of API keys, SSID etc.

Database: Database can be either local or in the cloud and stores the data generated by the IoT device.

Web Service: This acts as an interface between IoT device, application, database and analysis components. Web services can be implemented using HTTP and REST principle or using WebSocket protocol.

Analysis Component: The analysis component is responsible for analysing the IoT data and generates results which are easy for the user to understand. Analysis can be performed either locally or in the clouds.

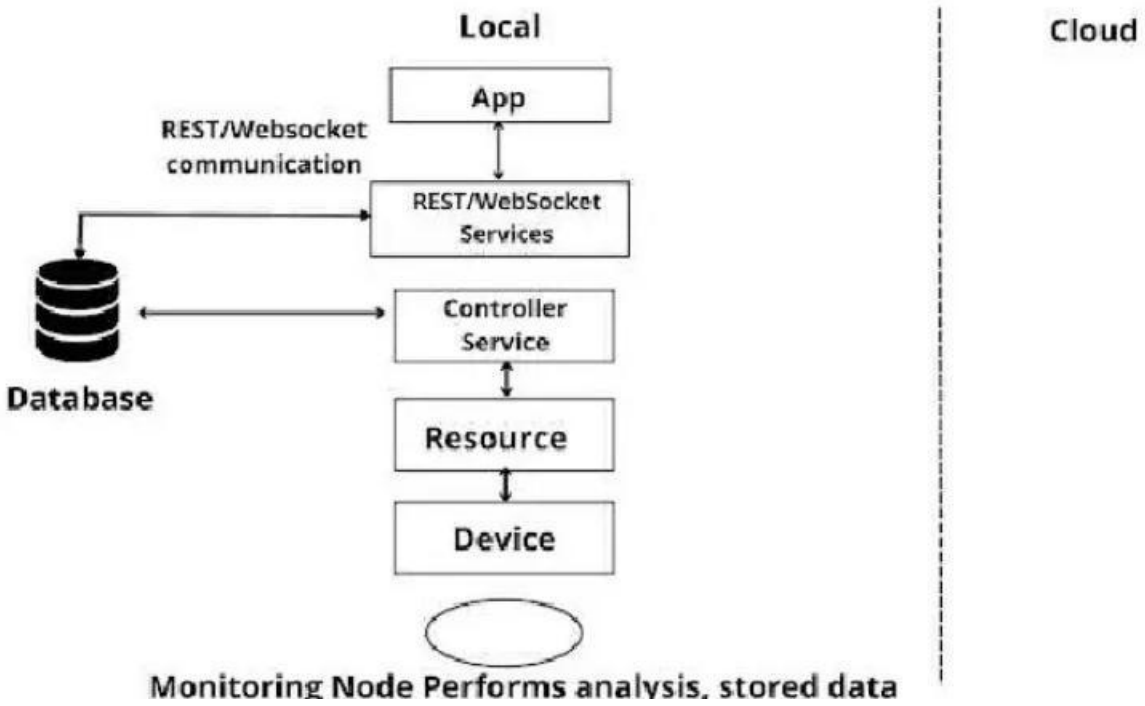
Application: IoT applications provide an interface that the user can use to control and monitor various aspects of the IoT system.

IoT level 1

IoT level has single node/device for sensing, monitoring, actuating, storing data, performing analysis and hosting application. Data involved is not big. So data is stored in local databases. It is suitable for modelling design low cost and low complexity solution. Primary analysis requirement is not computationally intensive. So can be analyzed locally.

EX: Home automation

Example: We can understand with the help of an example. Let's look at the IoT device that monitors the lights in a house. The lights are controlled through switches. The database has maintained the status of each light; in addition, REST services deployed locally allow retrieving and updating the state of each light and trigger the switches accordingly. To control the lights and applications, the application has an interface. The device is connected to the Internet and hence the application can also be accessed remotely

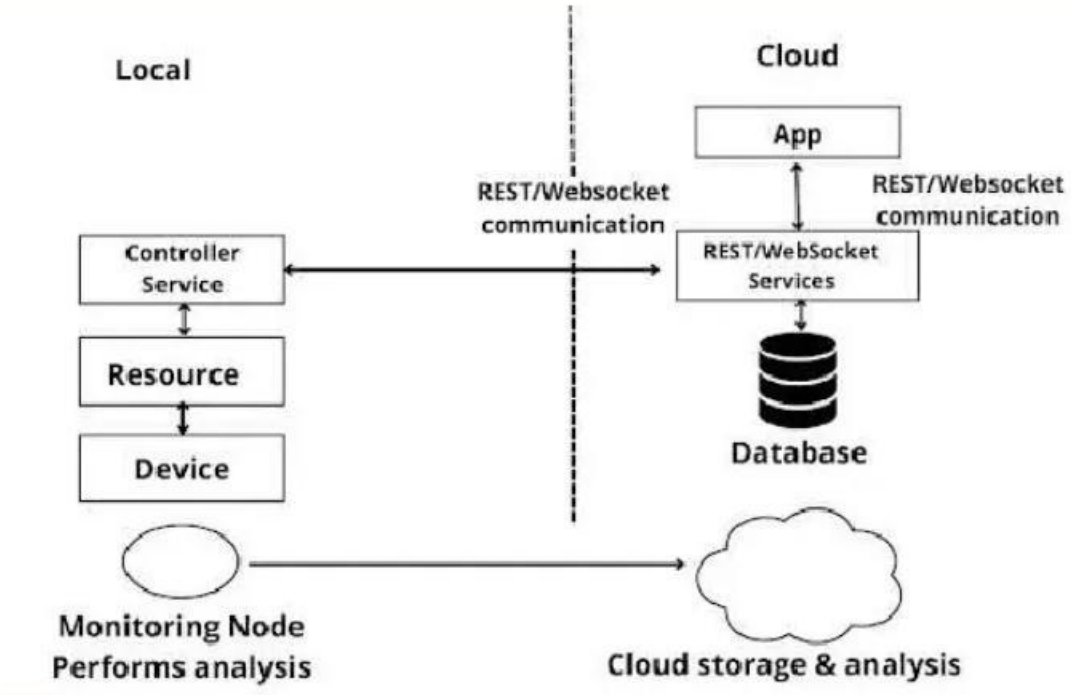


IoT level 2

IoT level 2 has single node/device for sensing, monitoring, actuating, performing analysis and hosting application. Data involved is big. So data is stored in clouds. It uses cloud-based application to visualize data. Primary analysis requirement is not computationally intensive. So can be analyzed locally.

EX: smart irrigation

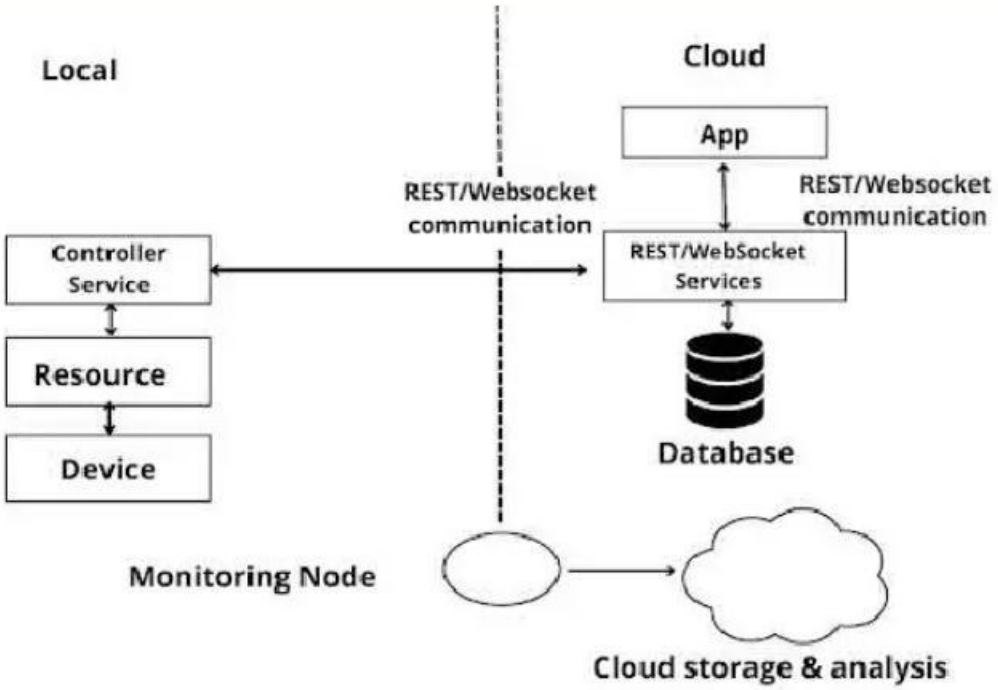
Example: A cloud-based application is used for monitoring and controlling the IoT system. A single node monitors the soil moisture in the field, which is sent to the database on the cloud using REST APIS. The controller service continuously monitors moisture levels



IoT level 3

level 3: It has single node/device for sensing, monitoring, actuating and hosting application. Data involved is big. So, data is stored in cloud. It uses cloud-based application to visualise data. Primary analysis requirement is computationally intensive. So can be aggregated and analyzed in cloud. Suchismita Mohanty

EX: tracking package handling.

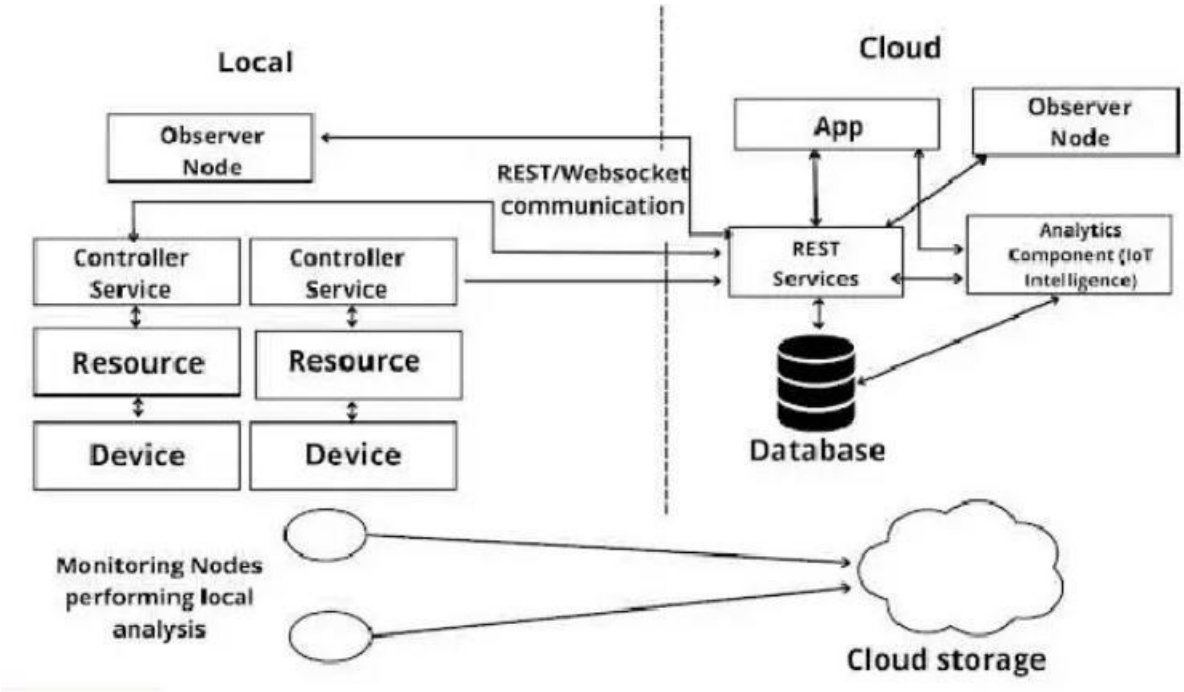


IoT level 4

IoT level 4 has multiple nodes/devices for sensing, monitoring, actuating, performing analysis and hosting application. Data involved is big. So data is stored in cloud. It uses cloud-based application to visualise data. Primary analysis requirement is computationally intensive. So can be aggregated and analysed in cloud.

It has two observer nodes **i.e local and cloud based**. They can subscribe to and receive information collected in cloud from IoT devices. They can process and use that information for various applications Observer node does not perform any control function.

EX: Noise monitoring

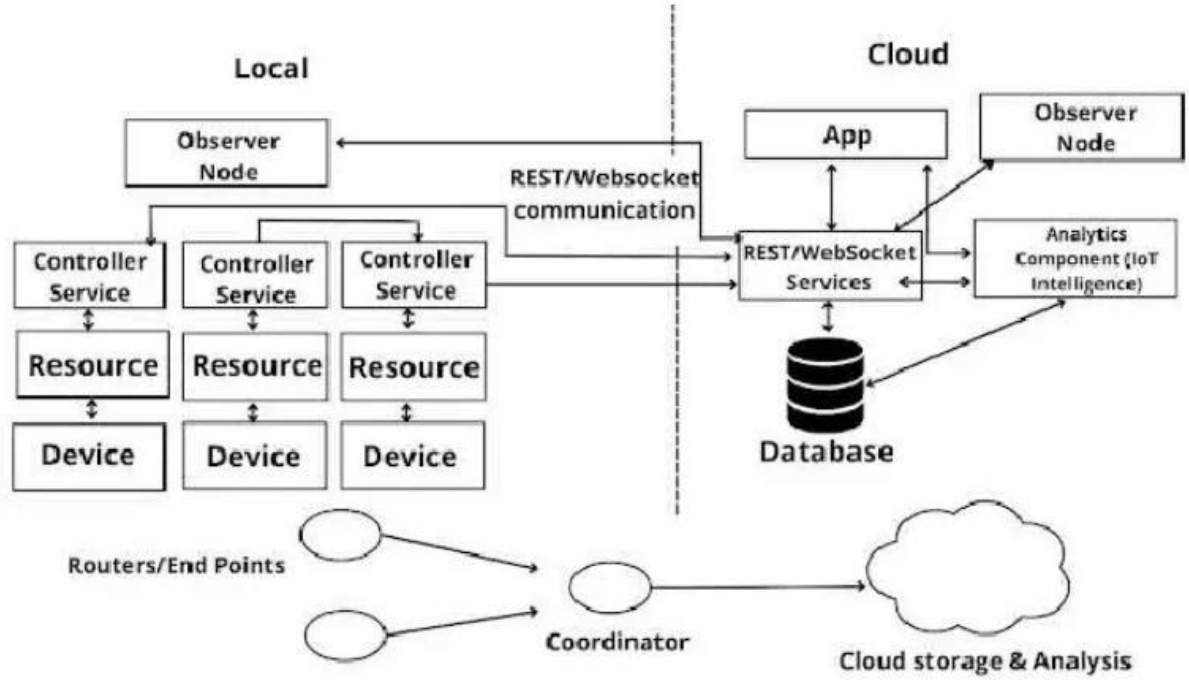

Local
Cloud
Observer Node
App
Observer Node
REST/Websocket communication
Controller Service
Controller Service
REST Services
Analytics Component (IoT Intelligence)
Resource
Resource
Device
Device
Database
Monitoring Nodes performing local analysis
Cloud storage


Integrated Applications
Information Processing
Network Infrastructure
Sensing and Identification

IoT level 5

IoT level 5 has multiple nodes/devices for sensing, monitoring, actuating. One coordinator node for collecting and sending the data to cloud by controller service. Data involved is big. So data is stored in cloud. It uses cloud-based application to visualise data. Suitable for wireless sensor networks. Primary analysis requirement is computationally intensive. Analytic components analyses the data and stores result in cloud and make predictions.

Ex: Forest Fire Detection

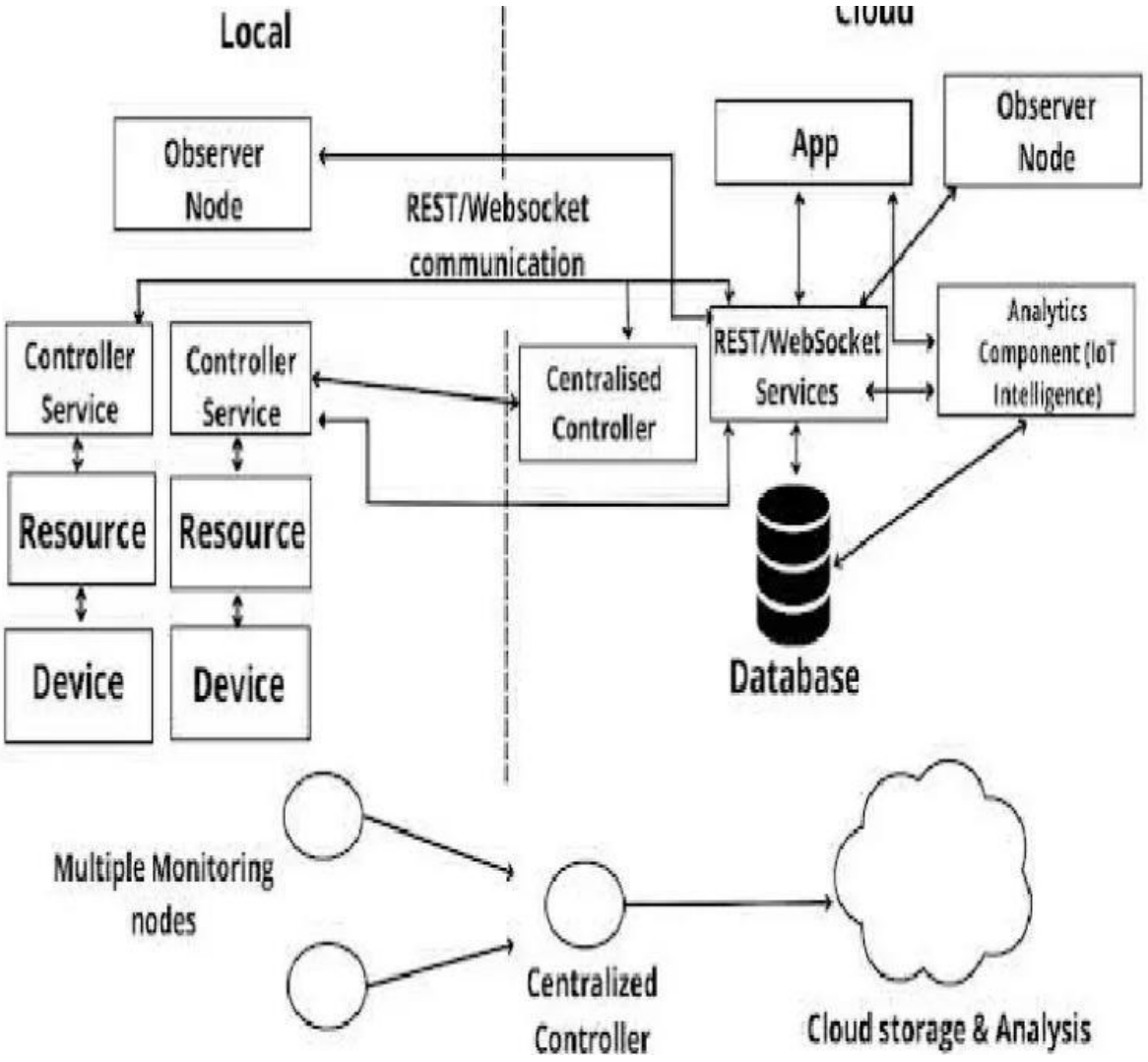


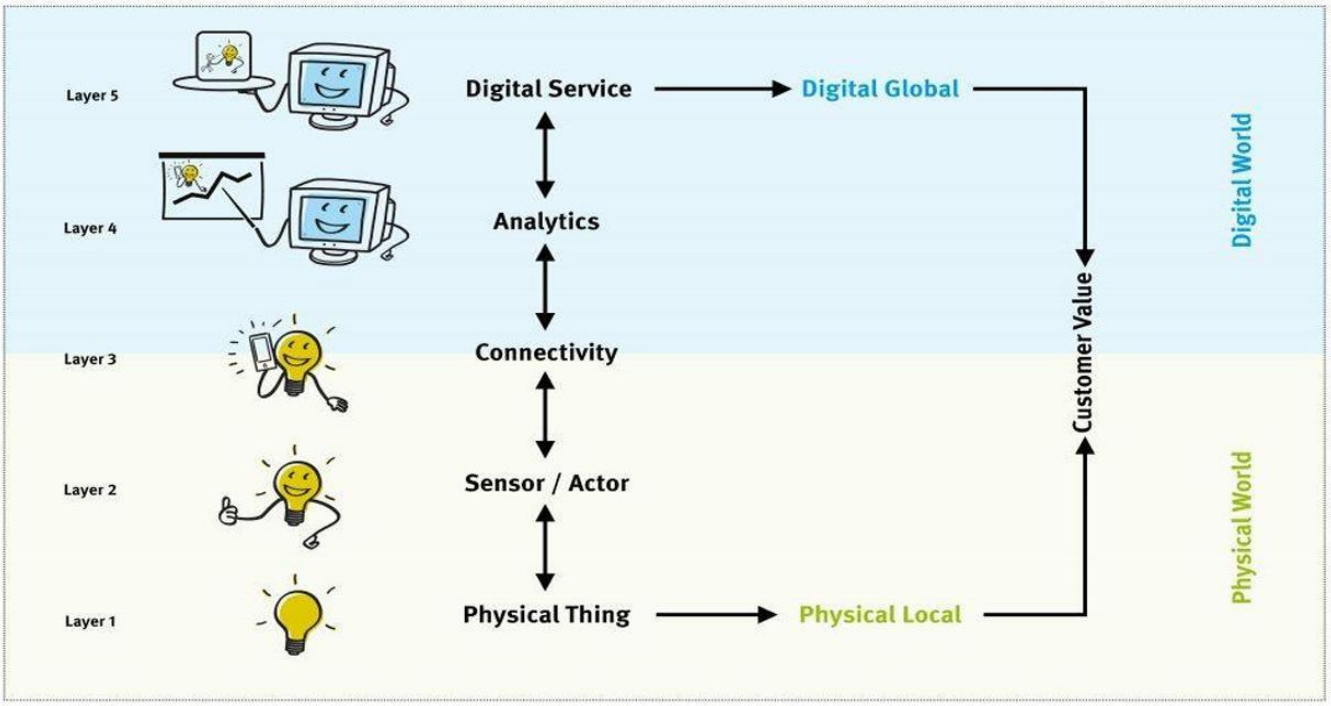
Example: A monitoring system has various components: end nodes collect various data from the environment and send it to the coordinator node. The coordinator node acts as a **gateway and** allows the data to be transferred to cloud storage using **REST API**. The controller service on the coordinator node sends data to the cloud

IoT level 6

IoT level 6 has multiple independent nodes/devices for sensing, monitoring, actuating and sending the data to cloud by controller service. Data involved is big. So data is stored in cloud. It uses cloud-based application to visualise data. Primary analysis requirement is computationally intensive. Analytic component analyse the data and stores result in clouds. It has centralized controller which is aware of the status of all the end nodes and sends control command to the nodes. Weather monitoring and structural health monitoring.

Example: Weather monitoring consists of sensors that monitor different aspects of the system. The end nodes send data to cloud storage. Analysis of components, applications, and storage areas in the cloud. The centralized controller controls all nodes and provides inputs




Layer 5
Layer 4
Layer 3
Layer 2
Layer 1
Digital Service
Analytics
Connectivity
Sensor / Actor
Physical Thing
Digital Global
Physical Local
Customer Value
Digital World
Physical World

IoT-ENABLING TECHNIQUES

IoT is enabled by several technologies, including wireless sensor networks, cloud computing, Big Data analytics, communication protocols, embedded systems etc.10. This section describes each and every enabling techniques used in IoT:

- **Wireless sensor network**
- **Cloud computing**
- **Big Data analytics**
- **Communications protocols**
- **Embedded system**

Wireless Sensor Network (WSN)

This comprises a number of distributed devices with sensors which are used to monitor the environmental and physical conditions. A wireless sensor network consists of end nodes, routers, and coordinators. End nodes have several sensors attached to them where the data is passed to a coordinator with the help of routers. The coordinator also acts as the gateway that connects WSN to the Internet.

Example:

- Weather monitoring system
- Indoor air quality monitoring system
- Soil moisture monitoring system

Cloud computing

Cloud computing provides us with the means by which we can access applications as utilities over the Internet. Using cloud computing, users can access any resources from anywhere like databases, web servers, storage, any device, and any software over the Internet.

It provides different services like:

- **IaaS (Infrastructure as a Service):** IaaS (Infrastructure as a Service) is

also known as Hardware as a Service (HaaS) It is one of the layers of the cloud computing platform and allows customers to outsource their IT infrastructures, such as servers, networking, processing, storage, virtual machines, and other resources. In traditional hosting services, IT infrastructure was rented out for a specific

- **PaaS (Platform as a Service):** Platform as a Service (PaaS) provides a

runtime environment. It allows programmers to easily create, test, run,

and deploy web applications. One can purchase these applications from a cloud service provider on a pay-per-use basis and access them using the Internet connection.

Example: Google App Engine, Force.com, Joyent, Azur

- **SaaS (Software as a Service):** This is also known as "On-Demand Software". It is a software distribution model in which services are hosted by a cloud service provider.

Example: Google Docs, Gmail, office

Big Data analytics

This refers to the method of studying massive volumes of data, or the socalled Big Data. It involves the collection of data whose volume, velocity, or variety is simply too massive and tough to store, control, process, and examine the data using traditional databases.

Communications protocols

They are the backbone of IoT systems and enable network connectivity and linking to applications. Communication protocols allow devices to exchange data over the network. Multiple protocols often describe different aspects of a single communication. A group of protocols designed to work together is known as a protocol suite

Embedded systems

It is a combination of hardware and software used to perform special tasks. It includes microcontroller and microprocessor memory, networking units (Ethernet Wi-Fi adapters), input output units (display keyword etc.), and storage devices (flash memory). It collects the data and sends it to the Internet. Embedded systems are used in a variety of applications:

1. Digital camera

2. DVD player, music player

3. Industrial robots

4. Wireless routers

IoT and M2M

M2M

Machine-to-Machine or M2M is a technology that allows connectivity between network devices. It allows tapping of sensor data and transmitting it over a public network. IoT technology, on the other hand, expands the concept of M2M by creating large networks of devices in which devices communicate with one another through cloud networking platforms. It allows users to create high performance, fast and flexible networks that can connect a variety of devices.

IoT and M2M

Various components that make up an M2M system are - sensors, **RFID (Radio Frequency Identification)** , Wi-Fi or cellular network, and a computing software which helps networking devices to interpret data and decision making. These M2M applications can translate data which in turn can trigger automated actions.

Various benefits offered by M2M are -

1. It reduces cost by making use of public network and minimizing downtime
2. Increase revenue by identifying new business opportunities
3. Increase customer satisfaction by timely servicing equipment and regularly monitoring.

M2M Architecture

M2M architecture consists of three domains:

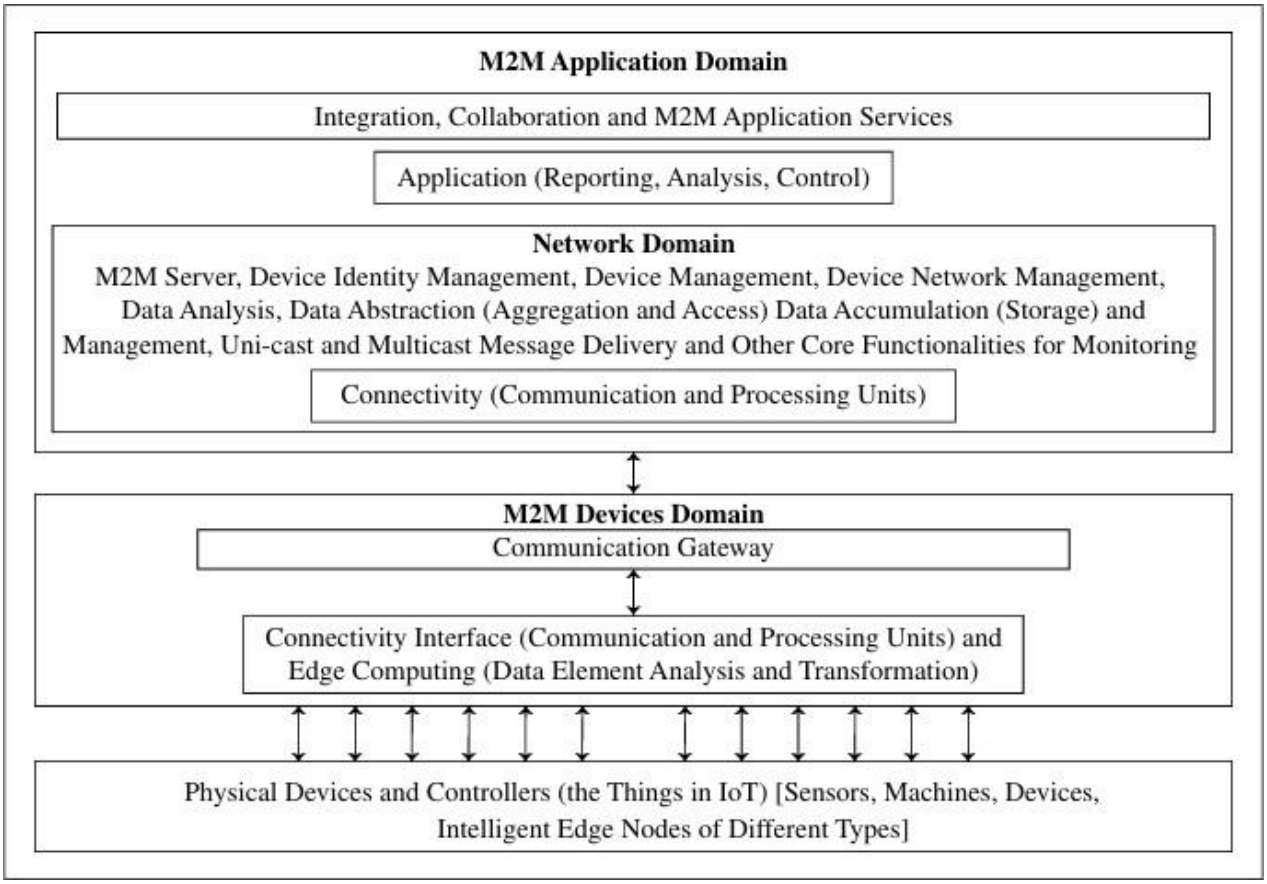
- M2M device domain
- M2M network domain
- M2M application domain.

M2M device communication domain consists of three entities: physical devices, communication interface and gateway. Communication interface is a port or a subsystem, which receives the input from one end and sends the data received to another.

M2M network domain consists of M2M server, device identity management, data analytics and data and device management similar to IoT architecture (connect + collect + assemble + analyse) level.

M2M application domain consists of application for services, monitoring, analysis and controlling of devices networks.

IoT and M2M



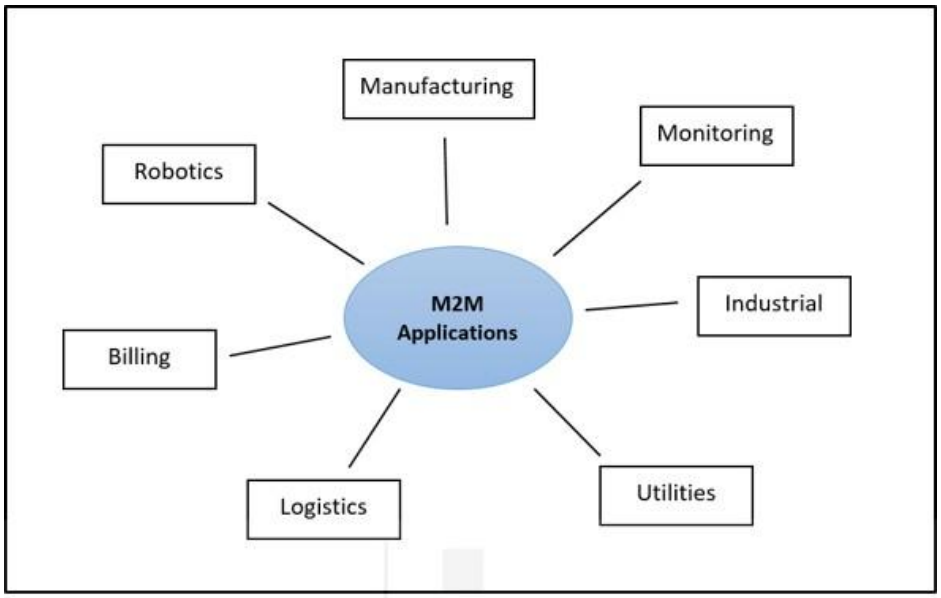
M2M Applications

Sensor telemetry is one of the first application of M2M communication. It has been used since the last century for transmitting operational data. Earlier people used telephone lines, then radio waves, to transmit measurements factors like- temperature, pressure for remote monitoring.

Another example of M2M communication is ATM. ATM machine routes information regarding request for transaction to appropriate bank. The bank in turn through its system approves it and allows transactions to complete.

It also has applications in supply chain management (SCM), warehouse management

systems (WMS), Utility companies, etc. Fig 1 shows various applications of M2M. Table 1 summarizes the differences between M2M and IoT devices.



IoT and M2M

M2M	IoT
1. M2M is subset of IoT	1. IoT is super set of M2M. It is a vast network of connected devices that may or may not include human interaction.
2. Point to point communication usually embedded within hardware at the customer site. Ex (Tap to pay, NFC). The service in Samsung taps to pay without using card. You just need to bring your mobile near the swipe machine. Using your ID card to open or close the door.	2. Multipoint Communication. Ex: By showing the ID card then your attendance will be registered. Monitoring the health parameters through the help of android app or web app.
3. It uses cellular or wired network.	3. It uses wireless network.
4. Do not necessarily require internet connection.	4. Majority of cases require Internetconnection.
5. It is more hardware based.	5. It is more software based.
6. On premises applications. (i.e software and technology i.e located within the physical confines of an enterprise or company's data centre.) Ex. Diagnosis applications, Service management applications and onpremises enterprises applications.	6 It uses cloud based application such as analytic applications, enterprise applications, and remote diagnosis and management applications.

7. Data collection and analysis is done in on premises storage infrastructure.	7. Data collection and analysis is done incloud mostly.
8. Non-IP based communication.	8. IP-based communication.
9. M2M protocols include ZigBee, Bluetooth, ModBus, M-Bus, Wireless M-Bus, Power Line communication (PLC), 6LoWPAN, IEEE 802.15.4, Z-wave. These protocols work below the network layer.	9. IoT protocols include HTTP, CoAP, Web Socket, MQTT, XMPP, DDS, AMQP etc.. The protocols work above the network layer.
10. The devices used are homogeneous type within an M2M area network.	10. The devices used are heterogeneous type such as fire alarms, door alarms, lighting control devices, etc

IoT systems can be implemented by four components.

1. **Sensors**

Sensors are devices that are capable of collecting data from the environment. There are various types of sensors available –temperature sensors, pressure sensors, RFID tags, light intensity detectors, electromagnetic sensors, etc.

2. **Network**

Data collected from sensors are passed over the network for computations to the cloud or processing nodes. Depending upon the scale, they may be connected over LAN, MAN or WAN. They can also be connected through wireless networks like- Bluetooth, ZigBee, Wi-Fi, etc.

3. **Analytics**

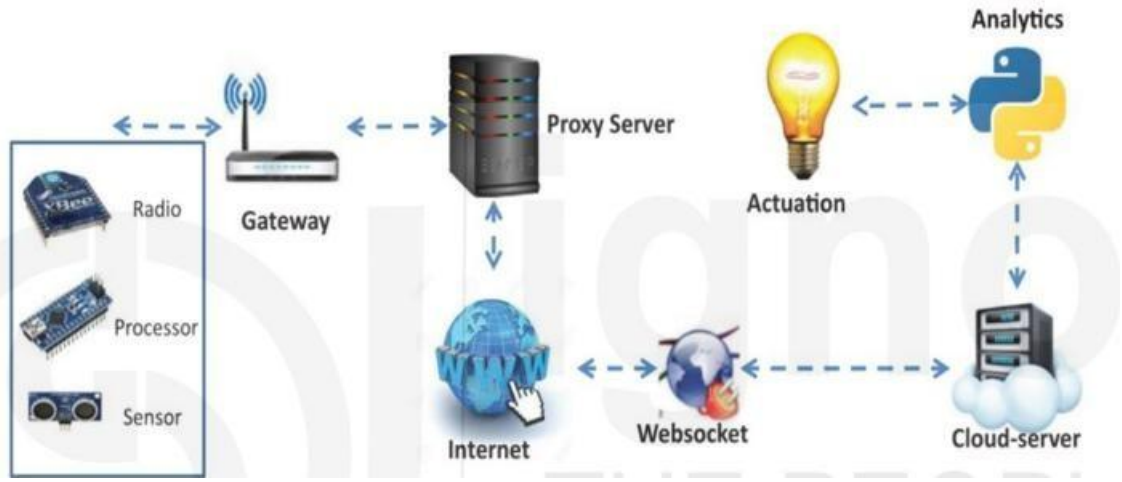
   The process of generating useful insights from the data collected by sensors is called analytics. Analytics when performed in real time, can have numerous applications and can make the IoT system efficient.

4. **Action**

   Information obtained after analytics must be either passed to the user using some user interface, messages, alerts, etc; or may also trigger some actions with the help of actuators. Actuators are the devices that perform some action depending on the command given to them over the network.

IoT and M2M

Fig shows implementation of IoT. Data captured by sensors are passed on to the cloud servers over the internet via gateways. Cloud servers in turn perform analytics and pass on the decisions or commands to actuators.



Gateway Prefix Allotment

Gateways are networking devices that connect IoT devices like sensors or controllers to Cloud. In other ways we can say that data generated by IoT devices are transferred to Cloud servers through IoT gateways.

The number of IoT devices is increasing at an exponential rate. These IoT devices are connected in a LAN or a WAN. A number of IoT devices within a building, communicating to a gateway installed in the same building over a wi-fi connection can be called an IoT LAN. Geographically distributed LAN segments are interconnected and connected to the internet via gateways to form IoT WAN. Devices connected within LAN have unique IP addresses but may have addresses the same as devices of another LAN.

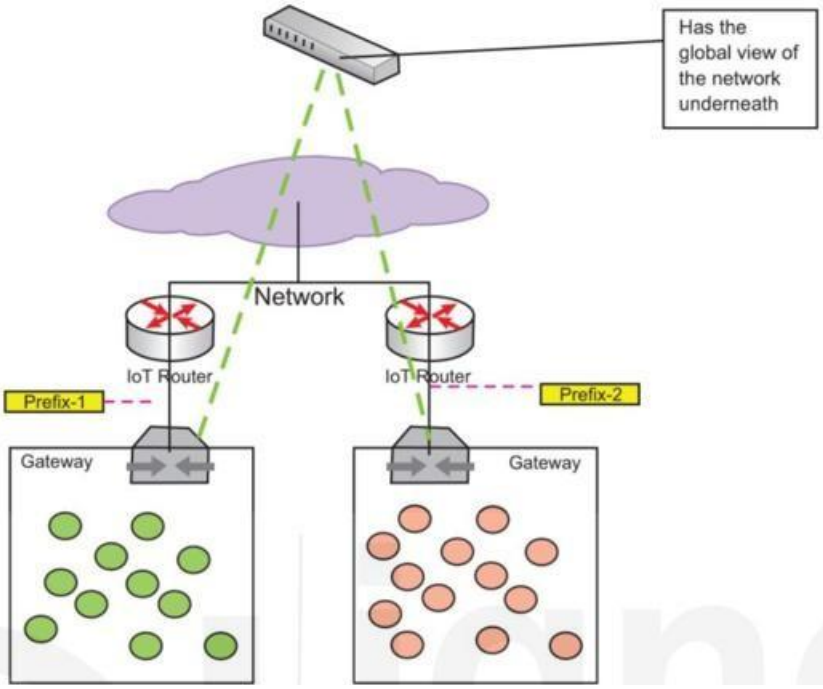
Gateways connect IoT LANs and WANs together. It is responsible for forwarding packets between them on the IP layer. Since a large number of devices are connected, address space needs to be conserved. Each connected device needs a unique address. IP addresses allocated to devices within a gateway's jurisdiction are valid only in its domain. Same addresses may be allocated in another gateway's domain. Hence to maintain uniqueness, each gateway is assigned a unique network prefix. It is used for global identification of gateways. This unique identifier removes the need of allocating a unique IP address to each and every device connected to the network, hence saves a lot of address space.

Gateway prefix allotment is shown in fig 3. Here two gateway domains are shown. Both of them are connected to the internet via router. This router has its own address space and allows connectivity to the internet. This router assigns a unique gateway prefix to both the gateways. Hence packets are forwarded from gateways to the internet via routers.

Impact Of Mobility on Addressing

When an IoT device moves from one location to another in a network, its address is affected. Network prefix allocated to gateways change due to mobility. WAN addresses allocated to devices through gateways changes without affecting IoT LAN addresses. This is possible because addresses allocated within a domain of gateway are unique. It is not affected by mobility of devices. These unique local addresses (ULA) are maintained independent of global addresses. For giving internet access to these ULAs, they are connected to application layer proxy which routes them globally.

Gateways are attached to a remote anchor point by using protocols like IPv6. These remote anchor points are immune to changes of network prefix. It is also possible for the nodes in a network to establish direct connection with remote anchor points to access the internet directly using tunneling. Fig 4 shows remote anchor points having access to gateways.



Data Management and Consolidation Gateway

Gateway includes the provisions for one or more of the following functions: transcoding and data management.

Following are data management and consolidation functions:

1. Transcoding
2. Privacy,
3. security Integration
4. Compaction and fusion

Transcoding

Transcoding does the data adaptation, conversion, and change of protocol, format or code Transcoding means data adaptation, conversion and change of protocol, format or code using software. The gateway renders the web response and messages in formats and representations required and acceptable at an IoT device. Similarly, the IoT device requests are adapted, converted and changed into required formats acceptable at the server by the transcoding software. For example, use of transcoding enables the message request characters to be in ASCII format at the device and in Unicode at the server. It also enables the use of XML format database at the device, while the server has a DB2, Oracle or any other database. Transcoding involves formats, data and code conversion from one end to another when the multimedia data is transferred from a server to the mobile TV, Internet TV, VoIP phone or smartphone as the client devices. Transcoding applications also involve filtering, compression or decompression. A transcoding proxy can execute itself on the client system or the application server. A transcoding proxy has conversion, computational and analysing capabilities, while a gateway has conversion and computational capabilities only.

Privacy

Data such as patient medical data, data for supplying goods in a company from and to different locations, and changes in inventories, may need privacy and protection from conscious or unconscious transfer to untrustworthy destinations using the Internet. Privacy is an aspect of data management and must be remembered while designing an application. The design should ensure privacy by ensuring that the data at the receiving end is considered anonymous from an individual or company.

Following are the components of the privacy model:

IoT and M2M

- Devices and applications identity-management
- Authentication
- Authorisation
- Trust Reputation

A suitable encryption of identification of data source enforces privacy.

Device ID management provides for privacy. The analysed decrypted data is an input to application, service or process. IoT or M2M data have to be for the beneficiary individual person or company only. When data is transfered from one point to another, it should be ensured that the stakeholder in future may not misuse the device end data or the application data. These static and dynamic relationships are components which depend on trust and reputation.

Secure Data Access

Access to data needs to be secure. The design ensures the authentication of a request for data and authorisation for accessing a response or service. It may also include auditing of requests and accesses of the responses for accountability in future. Example described how a layer provides the confidentiality and authorisation using AES-128 and CCM. End-to-end security is another aspect while implies using a security protocol at each layer, physical, logical link and transport layers during communication at both ends in a network.

Data Gathering and Enrichment

IoT and M2M

IoT/M2M applications involve actions such as data-gathering (acquisition), validation, storage, processing, reminiscence (retention) and analysis.

Data gathering refers to data acquisition from the devices/devices network. Four modes of gathering data are:

1. Polling refers to the data sought from a device by addressing the device; for example, waste container filling information in a waste management system

2. Event-based gathering refers to the data sought from the device on an event; for example, when the device reaches near an access point or a card reaches near the card reader or an initial data exchange for the setup of peer-to-peer or master-slave connection of BT device using NFC

3. Scheduled interval refers to the data sought from a device at select intervals; for example, data for ambient light condition in Internet of streetlights

4. Continuous monitoring refers to the data sought from a device continuously;

for example, data for traffic presence in a particular street ambient light condition in Internet of streetlights Data enrichment refers to adding value, security and usability of the data.

Data Dissemination

Consider the following three steps for data enrichment before the data disseminates to the network as aggregation, compaction and fusion.

Aggregation refers to the process of joining together present and previously received data frames after removing redundant or duplicate data.

Compaction means making information short without changing the meaning or context; for example, transmitting only the incremental data so that the information sent is short. **Fusion** means formatting the information received in parts through various data frames and several types of data (or data from several sources), removing redundancy in the received data and presenting the formatted information created from the information parts. **Data fusion** is used in cases when the individual records are not required and/or are not retrievable later.

Energy Dissipation in Data Dissemination

Energy consumption for data dissemination is an important consideration in many devices in WPANs and in wireless sensor nodes (WSNs). This is due to limited battery life. Energy is consumed when performing computations and transmissions. Higher the data rate, the greater will be the energy consumed. Higher is RF used, the greater will be the energy consumed. Higher the gathering interval, the lower will be the energy consumed. Energy efficient computations can be done by using concepts of data aggregation, compaction and fusion. Lesser the data bytes communication, greater the acquisition intervals, and lower the data rate for data transfer, lesser the energy dissipation.

Data Source and Data Destination

ID: Each device and each device resource is assigned an ID for specifying the data of source and a separate ID for data destination. Address: Header fields add the destination address (for example, 48-bit MAC address at Link layer, 32-bit IPv4 address at IP network and 128-bit IPv6 address at IPv6 network) and may also add the port (for example, port 80 for HTTP application).

Data Characteristics, Formats and Structures

Data characteristics can be in terms of temporal data (dependent on the time), spatial data (dependent on location), real-time data (generated continuously and acquired continuously at the same pace), real-world data (from physical world for example, traffic or streetlight, ambient condition), proprietary data (copy right data reserved for distribution to authorised enterprises) and big data (unstructured voluminous data). Data received from the devices, formats before transmission onto Internet.